Best Code + In-person Robot Practices

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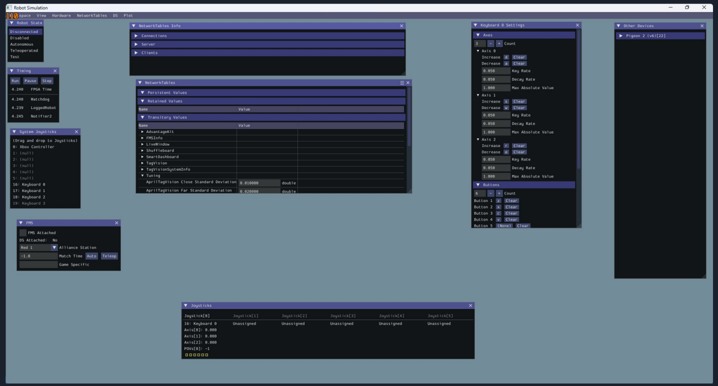
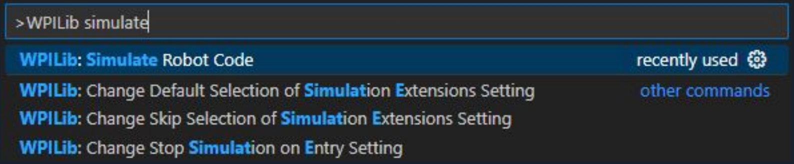
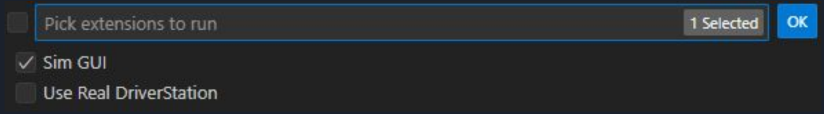
Agenda/Topics to Cover

- How to simulate robot code
- Using the simulation window
- What Advantage Scope is and how it works
- How to deploy robot code
- Driver station
- Using glass for tuning
- Capturing and downloading logs in Ascope

Simulating Robot Code





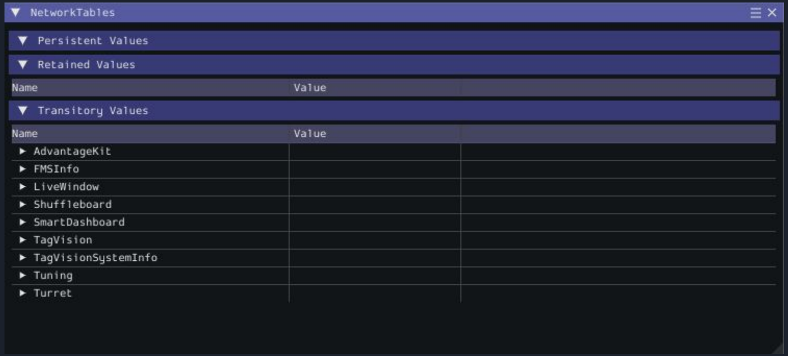


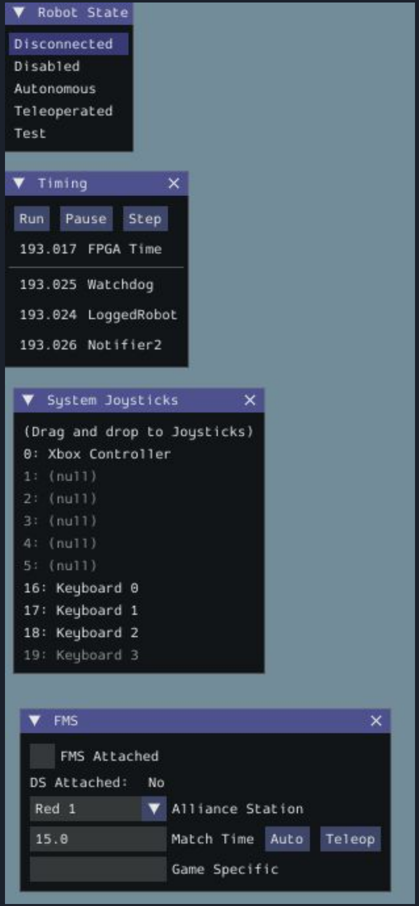
Issues You May Encounter

- Red Underlines
  - Import errors
  - Method, variable, etc. does not exist
- Yellow Underlines
  - Method, variable, etc. is not used

Simulation Window

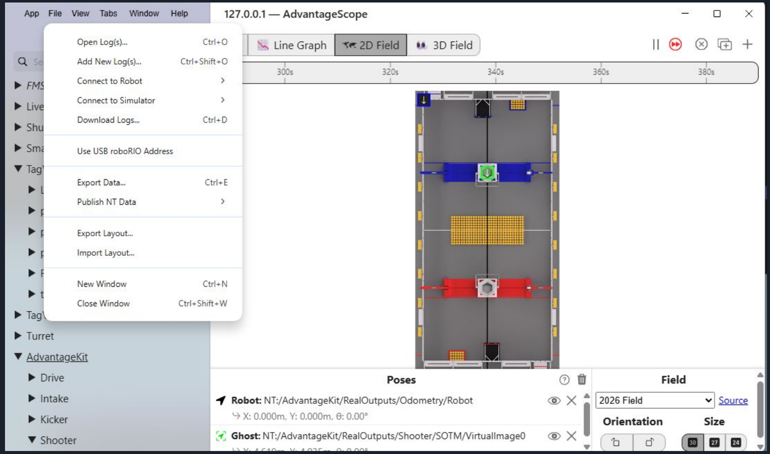
- Main interface for the simulation that automatically opens after simulating in vscode
- Contains some aspects that driver station includes, as well as some tuning capabilities





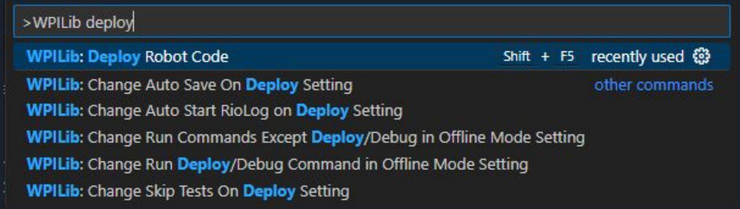
Introduction to Ascope

- Different ways to connect (under file tab in top left)
  - For simulation: ctrl-shift-k
  - For IRL: ctrl k
- Contains all kinds of information relating to the robot
  - In simulation, many values stay zero such as temperature since there's no way to know those simply through simulation
- Allows user to move around and control a virtual robot for testing purposes
  - Ex): checking if states update when buttons are pressed, how target pose updates, etc.



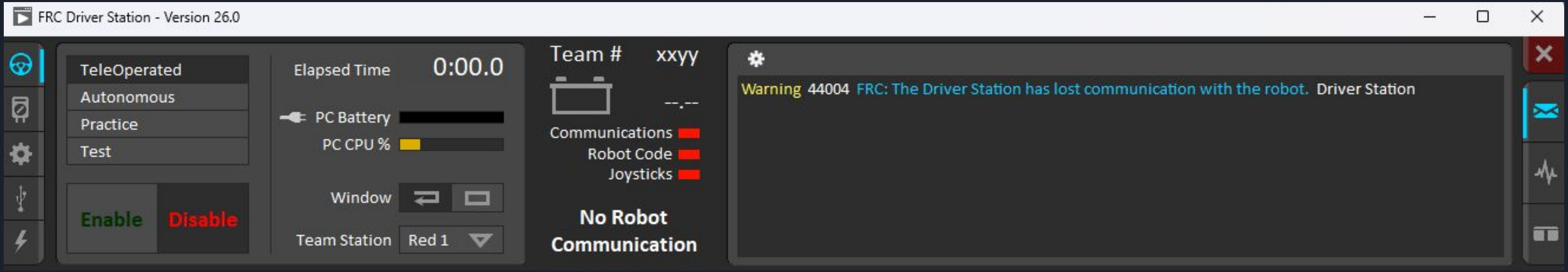
Deploying Robot Code

1. Connect to robot
   a. Direct connection by ethernet cable
   b. Connection over Wifi
2. Deploy code in vscode



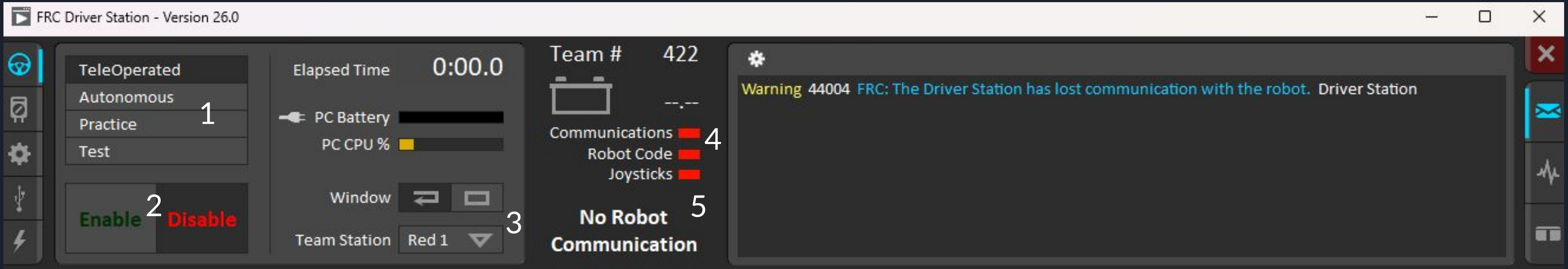
Driver Station

- Main interface for controlling the robot in real life
- Controls for robot
  - [ ] \ - press at the same time to enable robot
  - Enter - disables robot
  - Spacebar - E-Stop robot



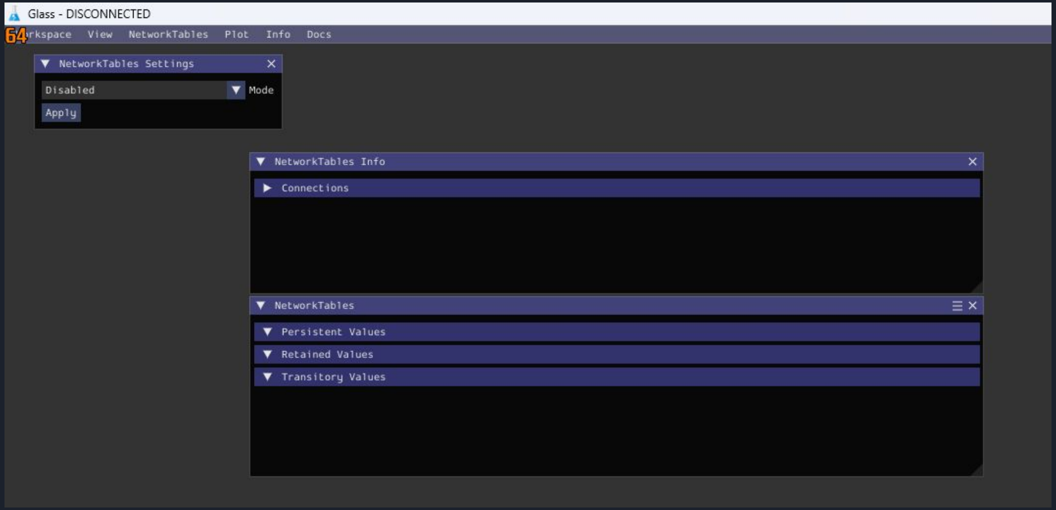
Navigating Driver Station

1. Adjusts mode of robot
2. Enable/disable robot
3. Opens a dropdown that allows user to select what station they're at
4. Shows status of communications, robot code, and joysticks. Turn green when driver station detects them and they work.
5. Status of robot (ex: teleoperated enabled/disabled, e-stopped)



Glass

- Software for tuning values
- When robot is connected, allows user to view all of the different values such as speeds, PID, etc. and adjust them as needed
- Changes in glass aren't permanent



Using and Downloading Logs in Ascope

- Allows you to track different values and states of the robot over time.
- Can be downloaded for future use/review.